

Student B – Assignment Submission

Q1. What does the coefficient of determination (R^2) measure, and what do values close to 1 versus close to 0 (or negative) tell you about the model?

Answer: R^2 tells you how good the regression is. The closer it is to 1, the better the model fits the data.

Q2. Compare Ridge, Lasso, and ElasticNet regularization. For each, state which norm of the weights it penalizes and its characteristic effect on the weights.

Answer: They all add a penalty term to shrink the weights so the model doesn't overfit. Lasso can push some weights to zero, so it makes the model sparse. ElasticNet is a combination of Ridge and Lasso.

Q3. What problem does RANSAC address in linear regression, and how does it work at a high level?

Answer: RANSAC is used when there are outliers in the data. It fits the line while ignoring the outliers so the result is more robust than a normal least-squares fit.

Q4. Why is logistic regression considered a classification method despite its name, and what is the role of the sigmoid (logistic) function?

Answer: It predicts a probability between 0 and 1 using the sigmoid function to squash the output, and then you threshold the probability to decide the class. So it is classification even though it is called regression.

Q5. In scikit-learn's LogisticRegression, what does the parameter C represent, and how do larger versus smaller values change the model?

Answer: C controls the regularization of the model.